SAFEMAN · STUDENT MODULE

SM-CSE

Confined Space & Gas Testing

SafeMan™ by Oryx Institute · safeman.oryxinstitute.org

Theory + 4-Gas Practicum · Pass mark 70% · 3 modules

Name _____

Student no. _____ Class _____



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WELCOME

SafeMan Confined Space Entry & Atmospheric Gas Testing

SafeMan Confined Space Entry & Atmospheric Gas Testing — Theory + 4-Gas Practicum. This SafeMan course is engineered for corporate audit readiness against: GN 156 Reg. 163; NTA Unit Std 848; ISO 45001 Cl. 8.1.

REGULATORY DISCLOSURE — READ FIRST

SafeMan™ is a proprietary inter-organizational safety management and operational competency benchmark developed by Oryx Institute (safeman.oryxinstitute.org). Courses are engineered in alignment with ISO 45001:2018, ISO 14001:2015, ISO 31000, ILO-OSH guidelines, and pre-referenced against Namibia Training Authority (NTA) Unit Standards and Namibian Government Notice 156 of 1997 for corporate audit readiness. SafeMan credentials verify industry competency and attendance between organizations and are non-NQA credit-bearing.

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How to use this module pack

Work through the modules in order. Each module states its focus and the key points you are expected to carry into practice. Diagrams referenced as web assets accompany the online reader at safeman.oryxinstitute.org. After the modules, sit the knowledge check: 10 questions, pass mark 70%, closed book.

This is a practicum course: after passing the knowledge check you must complete the assessor practicum checklist with a qualified assessor before your SafeMan credential is issued. A practical demonstration video slot is reserved in the online reader for every module.

MODULES

Module 1 — Confined spaces & their hazards

Definition first, then the hazard families that make confined spaces Namibia's silent killers.

Key points

A confined space is defined by access AND hazard potential: tanks, pits, silos, vessels, trenches, big pipe.

Atmospheric hazards: oxygen deficiency/enrichment, flammable vapours, toxic gases; plus engulfment, stored product, moving machinery.

Gases stratify: heavy sink (H2S, CO2), light rise (CH4) — test top, middle, bottom.

Rescue bodies from confined spaces outnumber rescuer losses when the attendant enters — the attendant never enters.

Diagram: /assets/webp/sm-gas-detector.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

Module 2 — Gas testing & the entry permit

4-gas detection discipline: bump tests, stratified sampling and the permit that gates every entry.

Key points

Acceptable entry atmosphere: O2 19.5–23.5%, LEL <10% (work), H2S and CO below their statutory/alarm limits.

Bump test with known gas before every use; calibration on schedule; zero in fresh air.

Sampling order: top, middle, bottom, at work level and remote corners; re-test on change.

Entry permit names the space, tests, isolations, ventilation, standby, rescue plan, duration — and is signed by all parties.

Diagram: /assets/webp/sm-gas-detector.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

Module 3 — Attendant duties & non-entry rescue

The attendant as the lifeline and the tripod-and-winch retrieval that keeps rescuers out of the tomb.

Key points

Attendant: constant watch, count and ID of entrants, no other duties, communication maintained, alarms on change.

Non-entry retrieval: tripod, winch, harness retrieval point rigged before entry.

Never enter for rescue without a rated atmosphere and backup — most double fatalities are would-be rescuers.

Ventilation continues while occupied; work stops when the alarm speaks.

Diagram: /assets/webp/sm-gas-detector.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

ASSESSMENT

Knowledge check

The closed-book knowledge check draws 10 questions from the bank reproduced in Appendix A. Pass mark 70%. Your SafeMan credential is issued after the knowledge check and the practicum sign-off, and remains valid for 24 months from issue.

Assessor Practicum Checklist — Confined Space Entry

Pass standard: Instrument verification, stratified sampling and permit completion to assessor standard.

1	4-gas detector zero and bump test verification performed and logged.	
2	Pre-entry stratified sampling sequence (top/middle/bottom) executed with correct readings recorded.	
3	Confined Space Entry Permit completed with isolations, ventilation and standby named.	
4	Attendant watch established with communication and non-entry retrieval rig checked.	
5	Permit lock-off and close-out demonstrated.	

Assessor name, role/qualification and company are captured on the sign-off screen of the SafeMan platform and bound into the QR-verified certificate record.

APPENDIX A — QUESTION BANK

Knowledge check bank

Q1. Acceptable oxygen range for entry is:

A. 16–25%

B. 19.5–23.5%

C. 18–21%

D. 20–28%

Q2. LEL limit for work in a confined space:

A. Below 50%

B. Below 10%

C. Below 25%

D. Zero forever

Q3. Gases stratify, so testing is done:

A. At the entrance only

B. Top, middle and bottom

C. Once per shift

D. By smell

Q4. H2S is:

A. Lighter than air and sweet

B. Heavier than air and deadly at low ppm

C. Only a mine problem

D. Detectable by smell at all levels

Q5. A bump test verifies:

A. The casing

B. The sensors respond to known gas

C. Battery only

D. The pump

Q6. The attendant's rule is:

A. Enter to rescue immediately

B. Never enter — maintain watch and summon rescue

C. Leave for coffee

D. Double as gas tester

Q7. An entry permit records:

A. The tea roster

B. Tests, isolations, ventilation, standby, rescue plan and duration

C. Only names

D. The client invoice

Q8. Non-entry retrieval equipment:

A. Rope from a bakkie

B. Tripod, winch and retrieval harness rigged before entry

C. A ladder

D. A fire hose

Q9. Ventilation during occupancy:

A. Stops at entry

B. Continues while occupied

C. Only after alarms

D. Is optional

Q10. Atmospheric change mid-task means:

A. Finish quickly

B. Stop work and exit, re-test

C. Report at shift end

D. Work faster

Answer key

Q1=B Q2=B Q3=B Q4=B Q5=B Q6=B
Q7=B Q8=B Q9=B Q10=B